Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Average velocity

Meaning in my words How would you explain average velocity to a friend?	**How can you tell?** A cart travels out and back. Which two points determine its average velocity?
Example or sketch Right is positive. A cart goes from +8 m to +2 m in 3 s. What is its average velocity?	**Non-example and why** “Average the two speeds, and you always get average velocity.” What is missing?

Average speed

Meaning in my words How would you explain average speed to a friend?	**How can you tell?** On a trip out and back, which distance goes on top of the fraction?
Example or sketch You walk 3 m out and 3 m back in 4 s. What is your average speed?	**Non-example and why** “I got back to the start, so my average speed was zero.” What would you say?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slope

Meaning in my words How would you explain slope to a friend?	**How can you tell?** On a position–time graph, what does a line falling to the right tell you?
Example or sketch A line goes from 2 m at 0 s to 8 m at 3 s. What slope would you label?	**Non-example and why** The graph reaches 8 m. Does that make the slope 8 m/s?

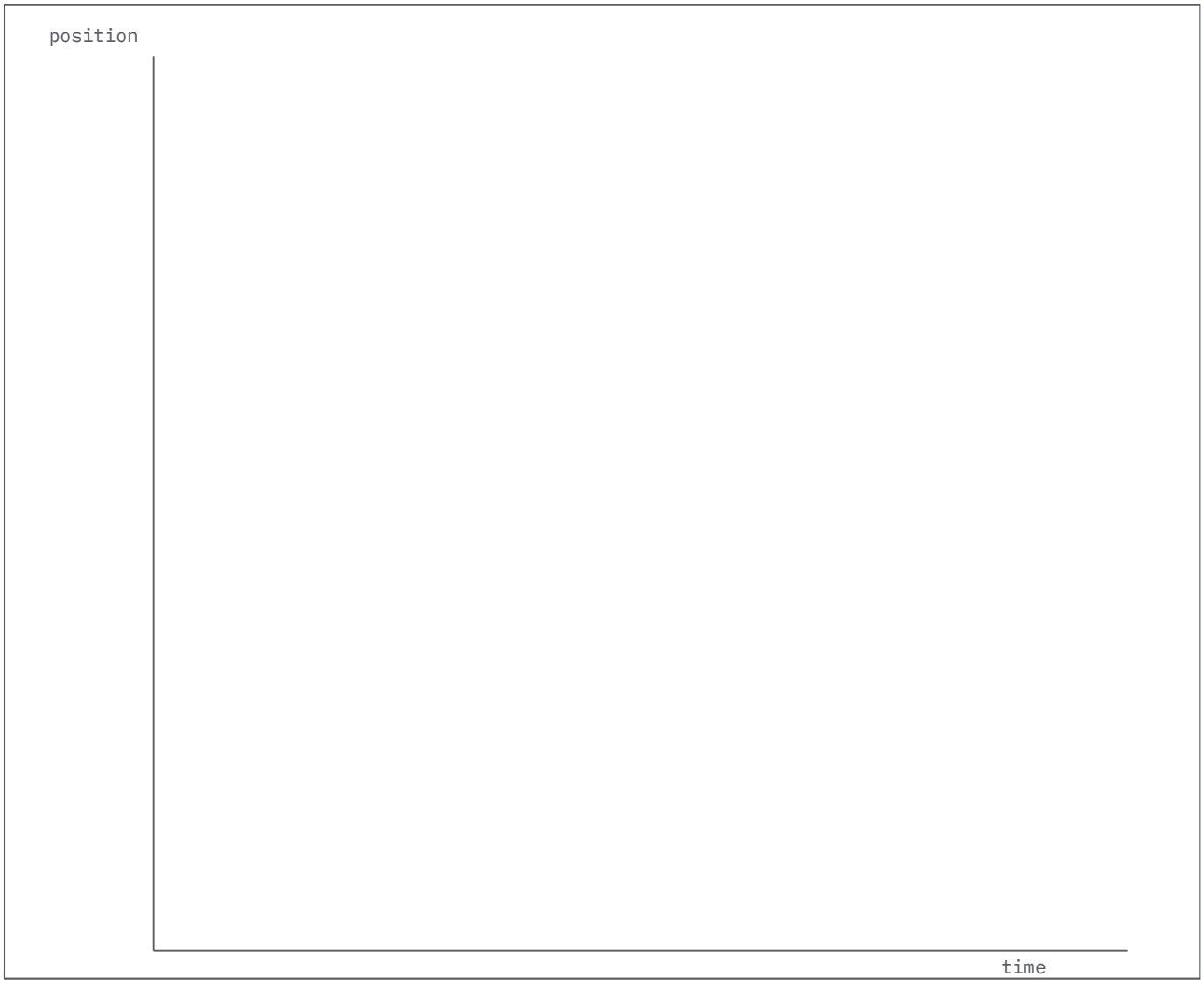
Instantaneous velocity

Meaning in my words How would you explain instantaneous velocity to a friend?	**How can you tell?** On a curved position–time graph, what line would you draw at exactly 2 s?
Example or sketch What does a tangent slope of −3 m/s at 2 s tell the camera operator?	**Non-example and why** A partner joins the first and last points of a curved trip. Which velocity does that slope give?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a falling position–time line. Label the signed rise, positive run and velocity unit.



My marks mean

On assignment: Average velocity

SECTION E / BACK / FINAL MEETING ONLY Name _____

Radar sees XJ at +12 km, then +6 km three seconds later. Positive points away from Earth.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Radar operator**

Send a signed velocity to the tracking console. Mark the two fixes on a sloping line so the next operator can check the sign.

☐ **Traffic broadcaster**

Give viewers a speed and direction. Your producer points at the two readings and asks whether they prove XJ is slowing.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor's check: Say what two sightings cannot establish about the velocity between them.